



Intaliq

SE322: Software Design and Architecture

Section: 1340

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1. Introduction

1.1 Overview

Intaliq aims to provide a solution to multiple issues Saudi Arabians are facing with outdoor activities currently, including the following:

- Monitoring outdoor activity: There is a need for fitness tools that accurately monitor users' outdoor activity levels.
- Finding partners: A lot of people struggle with finding partners to join them in their preferred outdoor activity.
- Joining Group Sessions: It is hard to find big groups for larger scale activity events that are led by professionals in the field especially in Saudi Arabia.

The Kingdom of Saudi Arabia is undergoing a bold transformation as part of Vision 2030, which aims to strengthen the quality of life with a specific emphasis on well-being and health of its citizens. The government has set visionary goals, one of the main goals being to increase the life expectancy average from 74 years to 80 years and significantly jump in the population's rate of weekly exercise from 13% to 40%. These goals highlight the importance of physical activity as a key factor in reducing chronic diseases. This is where our app Intaliq comes into play, empowering citizens to adopt a more active and healthier lifestyle.

1.2 Purpose

Intaliq is a fitness mobile application created to make outdoor activities, such as running, cycling, and hiking, more accessible and enjoyable in Saudi Arabia. It allows users to track their daily activity level, set goals for themselves, and join other users in group sessions. This software helps guide users through outdoor activities by providing a trained team of professionals that hold daily classes. Intaliq has a simple intuitive design that makes it easy for users to navigate routinely, as well as aids outdoor activities appear less intimidating.

The primary objectives of this system are:

- Assist users with navigating outdoor activities in Saudi Arabia by delivering an easy-to-use interface.
- Provide accurate tools for monitoring and setting activity goals.

- Help users make social connections and build relationships by creating a platform that connects people all over Saudi Arabia who have an interest in outdoor activity.

By combining these features into a single app, Intaliq promotes not only personal wellness but also a healthier society and community. It supports the government objectives to promote proactive and preventive healthcare practices, encourages the population in regular physical activities and exercise, and creates a community that values and embraces fitness and well-being . Motivated by Vision 2030, Intaliq strives to play a significant role in evolving and turning Saudi Arabia into a healthier, more active society with an improved lifespan and well-being.

1.3 Scope

The Intaliq system is a fitness software for smart phones that was created to promote outdoor exercises in Saudi Arabia. Users are able to log activities such as running, cycling, and hiking, set personal fitness goals, and take part in fitness coach-led group sessions.

Through social connections and organized activities, the system seeks to increase the accessibility, motivation, and social engagement of outdoor fitness.

Intaliq has a variety of features that are built to aid Saudi citizens in navigating outdoor activities. Some of the key features include:

- Activity Tracking: The system allows users to log their daily outdoor activities and track improvements.
- Activity Goal Setting: Users can set and adjust their activity goals to monitor progress.
- AI Analysis: Users can get an overview of their improvements by chatting with an AI chatbot or make adjustments to their goals.
- Finding Partners: The system provides a feature that allows users to chat and find partners to help motivate each other.
- Joining large groups for sessions: Users can locate and join a big group in large scale sessions, such as: running crew, hiking or cycling groups, hosted by Intaliq's own fitness team.
- Creating Group Sessions: Coaches from Intaliq's fitness team can create and schedule a new group session for users to join.

- Admitting Group Members: A Coach can accept users' applications to join sessions, and admit them if their goals and current level matches with the session's objectives.
- Announcing Session Updates: The system allows coaches to make announcements about their scheduled sessions ahead of time, and any changes that may occur.

1.4 Definitions, acronyms, and abbreviations

Session class type		
Data Name	Data type	Constraint
date	string	- By default, must follow the template: DD/MM/YYYY - Can't be an unrealistic date (far future/past)
time	string	- By default, must use the 24-hour clock time
location	string	- Must be located in Saudi Arabia - Must be permitted for group physical activities
capacity	int	- Must be more than 1 - Depending on event type and scale, location must accomodate
intensity	string	- Must fall under beginner, intermediate, or advanced

User class type		
Data Name	Data type	Constraint
firstName	string	- Must be less than 20 characters
lastName	string	- Must be less than 35 characters
phoneNumber	string	- Saudi Arabian phone number must follow the template: +966 5X-XXX-XXXX - Non-Saudi Arabian phone numbers must have their country code followed by their number
email	string	- Domain must be officially recognized - Email handle must already exist - Email handle should be less than 64 characters
password	string	- Must be 8 characters minimum with 1 upper case, 1 number, and 1 special character

Coach class type		
Data Name	Data type	Constraint
firstName	string	- Must be less than 20 characters
lastName	string	- Must be less than 20 characters
phoneNumber	string	- Saudi Arabian phone number must follow the template: +966 5X-XXX-XXXX - Non-Saudi Arabian phone numbers must have their country code followed by their number
email	string	- Domain must be officially recognized - Email handle must already exist - Email handle should be less than 64 characters
password	string	- Must be 8 characters minimum with 1 upper case, 1 number, and 1 special character

Goal class type		
Data Name	Data type	Constraint
type	string	- Follows the 3 activities that Intaliq focuses on: Hiking, Biking, and Running
targetTime	string	- Must follow template HH:MM
caloriesBurnt	string	- Must be more than 10 kCal
deadline	string	- By default, must follow the template: DD/MM/YYYY - Can't be an unrealistic date (past)

Partner class type		
Data Name	Data type	Constraint
location	string	- Must be located in Saudi Arabia - Must be permitted for group physical activities
goals	string	- Must relate to the 3 activities that Intaliq focuses on: Hiking, Biking, and Running - Other constraints mentioned in goal class must be noted

Partner class type		
activities	string	- Follows the 3 activities that Intaliq focuses on: Hiking, Biking, and Running

Activity class type		
Data Name	Data type	Constraint
type	string	- Follows the 3 activities that Intaliq focuses on: Hiking, Biking, and Running
duration	string	- Must be 5 minutes and more - Shouldn't exceed a day

1.5 References

- Intaliq Vision and Scope Document
- IEEE Software Requirements Specification (SRS) Template

2. Architectural Representations

We use the 4+1 View Model, introduced by Philippe Kruchten in 1995, to describe the architecture of Intaliq. Since Intaliq has different parts that need to be considered from different perspectives, we used the model to break the architecture into five related views. The system is mainly used by regular users and coaches, while developers are responsible for building and maintaining the application. Looking at the system from these different perspectives makes it easier to explain how its components work together. The model consists of four main views: Logical, Process, Development, and Physical, along with the User View (the “+1”). The User View is based on use cases and scenarios and helps connect the other views to the way users interact with the system. The following sections describe each view and how it applies to Intaliq. Using the 4+1 View Model also gives us a clear way to organize the architecture and show how the different parts of Intaliq fit together.

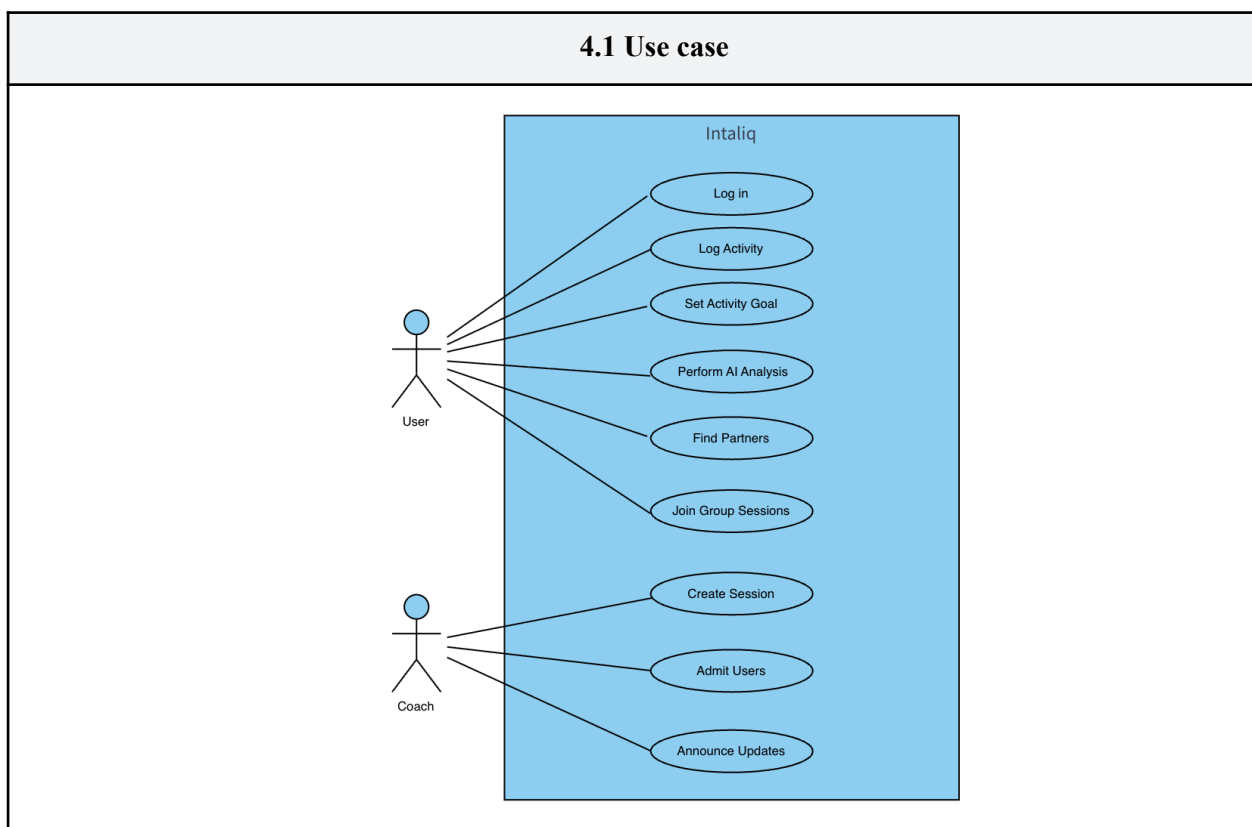
3. Architectural Goals and Constraints

The architecture of intaliq has goals that align with its purpose of encouraging Saudi citizens to be physically active and connect with others through sports. Some of the goals associated with the architecture are:

- providing an easy-to-use interface on mobile phones
- the protection of user accounts and keeping their information safe, including its integrity and confidentiality.

Also, some constraints will limit the implementation and architecture of the app, including the language. In the project, JavaScript will be used to develop the application. Other limitations, like time and cost for this project, will be discussed later

4. User View



4.2 Scenarios

4.2.1 Book a session	
Actors	Users

4.2.1 Book a session	
Description	A user can browse the available group sessions and book a session to take part in outdoor exercises, such as cycling, hiking, or running. The user can view available sessions, check session details, including the coach, and submit a request to join the session.
Data	User information. Session details: activity type, time, date, location, capacity of a session, intensity level of a session, and coach information.
Stimulus	User command on application selects and books a session
Response	Confirmation message that the booking request has been submitted. If accepted by the coach, the user is added to the session. If rejected, the user is notified.

Normal Scenario:

1. User opens the Intaliq app
2. User logs into the system
3. User browses available group sessions
4. User select session
5. User the review session details and coach details
6. User submits a booking request
7. The coach reviews the request
8. The coach accepts the request
9. The system confirms the booking and notifies the user

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message
- If the session is full —> the system prevents sending the request with a pop-up message
- If the coach rejects the request —> the system notifies the user
- If a system error occurs —> the request is not completed and the user is notified

4.2.2 Communicate with AI Chatbox	
Actors	Users
Description	A user can chat with an AI Chatbox for fitness related advice.
Data	User entered-query
Stimulus	User command on AI chatbox chat
Response	AI Chatbox will answer user accordingly and suggest further help.

Normal Scenario:

- 1- User opens the Intaliq app
- 2- User logs into the system
- 3- User selects more tab
- 4- User select AI Chatbox
- 5- User select from given chat options or write their own
- 6- User submits the chat option by clicking
- 7- AI coach will answer the chosen query
- 8- User accepts answer and closes chat

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message
- If user asks for medical advice that should be only given from trained specialists —> AI chatbox can refuse answer and end chat
- If a system error occurs —> the chat request is not completed and the user is notified

4.2.3 Enter activity goal	
Actors	Users.
Description	A user can enter an activity goal they want to achieve.
Data	User information, activity type, specific goal to achieve.
Stimulus	User command on application.
Response	Confirmation by the system that the goal has been saved.

Normal scenario:

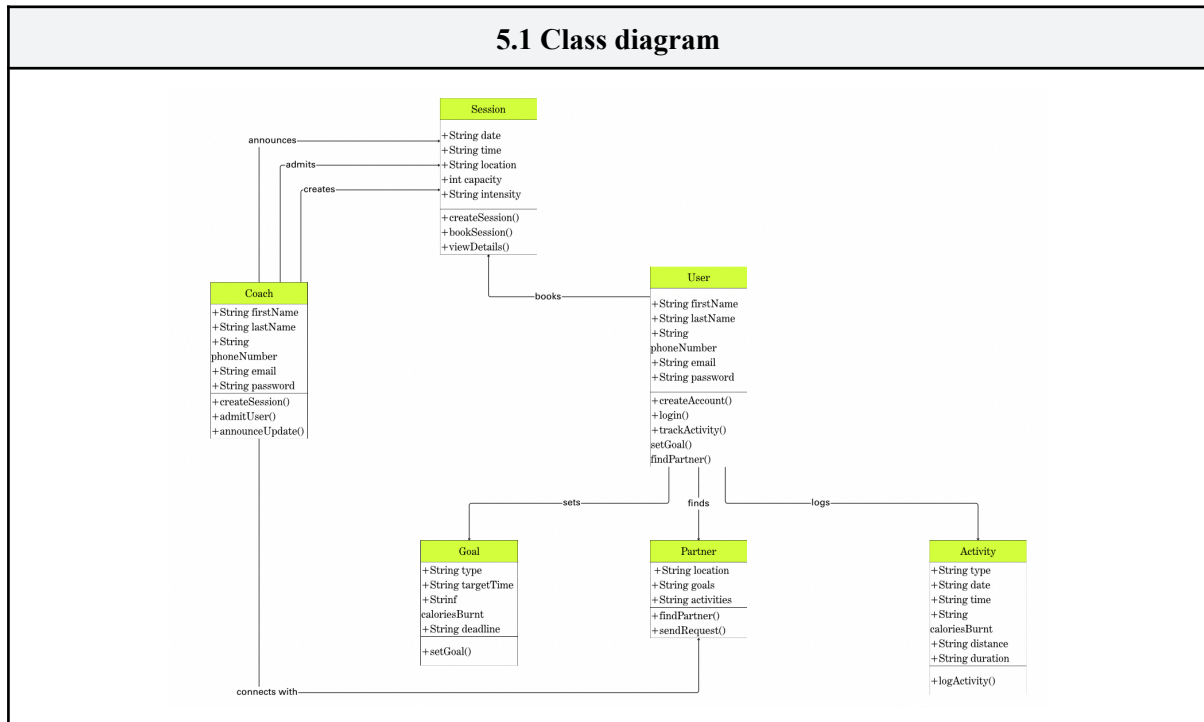
- 1- The user opens the Intaliq app.
- 2- The user logs into the system.
- 3- The user selects “Activity”.
- 5- The user selects “Activity Goal”.
- 4- The user enters their activity goal (activity type, goal).
- 5- The user submits the goal.
- 6- The system confirms the input.
- 7- The system saves the goal.
- 8- The system confirms the goal has been successfully created.

Alternative Scenario:

- If the user is not logged in —> system prompt login with a pop up message

- If any required field is missing —> The system prompts the user to fill in the missing information.
- If invalid data is entered —> The system displays an error message.
- If a system error occurs —> The goal is not saved.

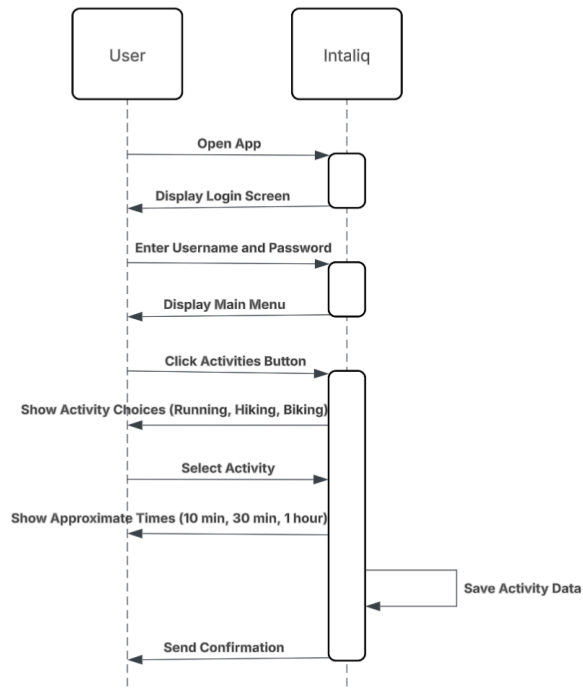
5. Logical View



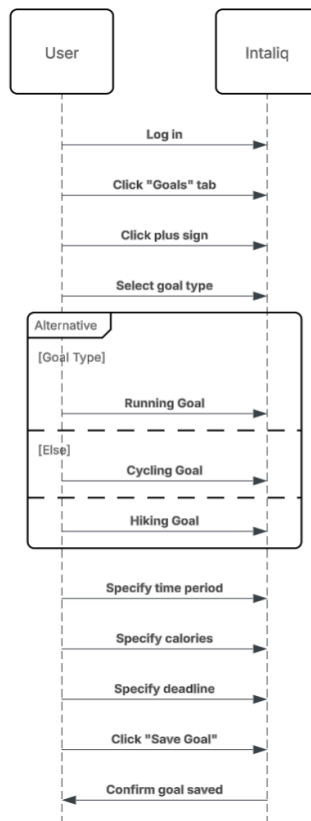
Our class diagram includes the following classes: Session, User, Goal, Partner, Activity, and Coach. Users can start goals, and log their activities. They can also find partners, and even book sessions created by coaches. Coaches, based on their judgement, can either admit or reject users into a session. To get more awareness, coaches can announce sessions as well. Lastly, coaches can connect with partners.

5.2 Sequence Diagrams

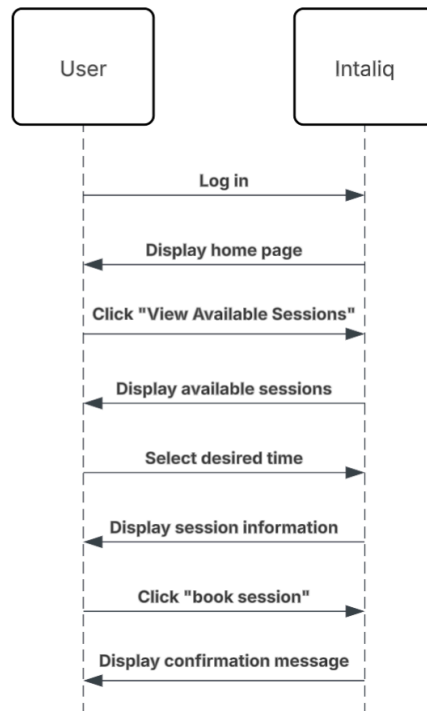
Logging an Activity



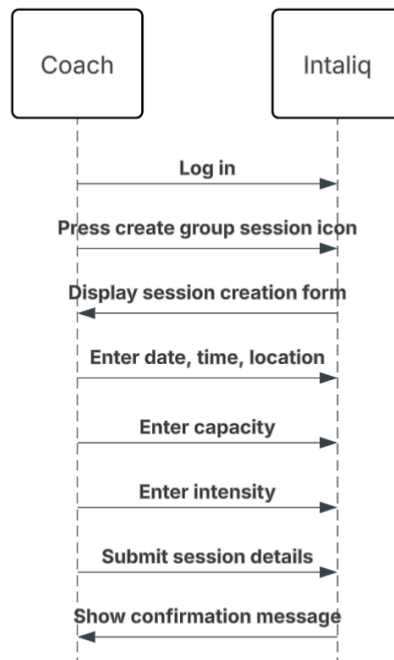
Setting an Activity Goal

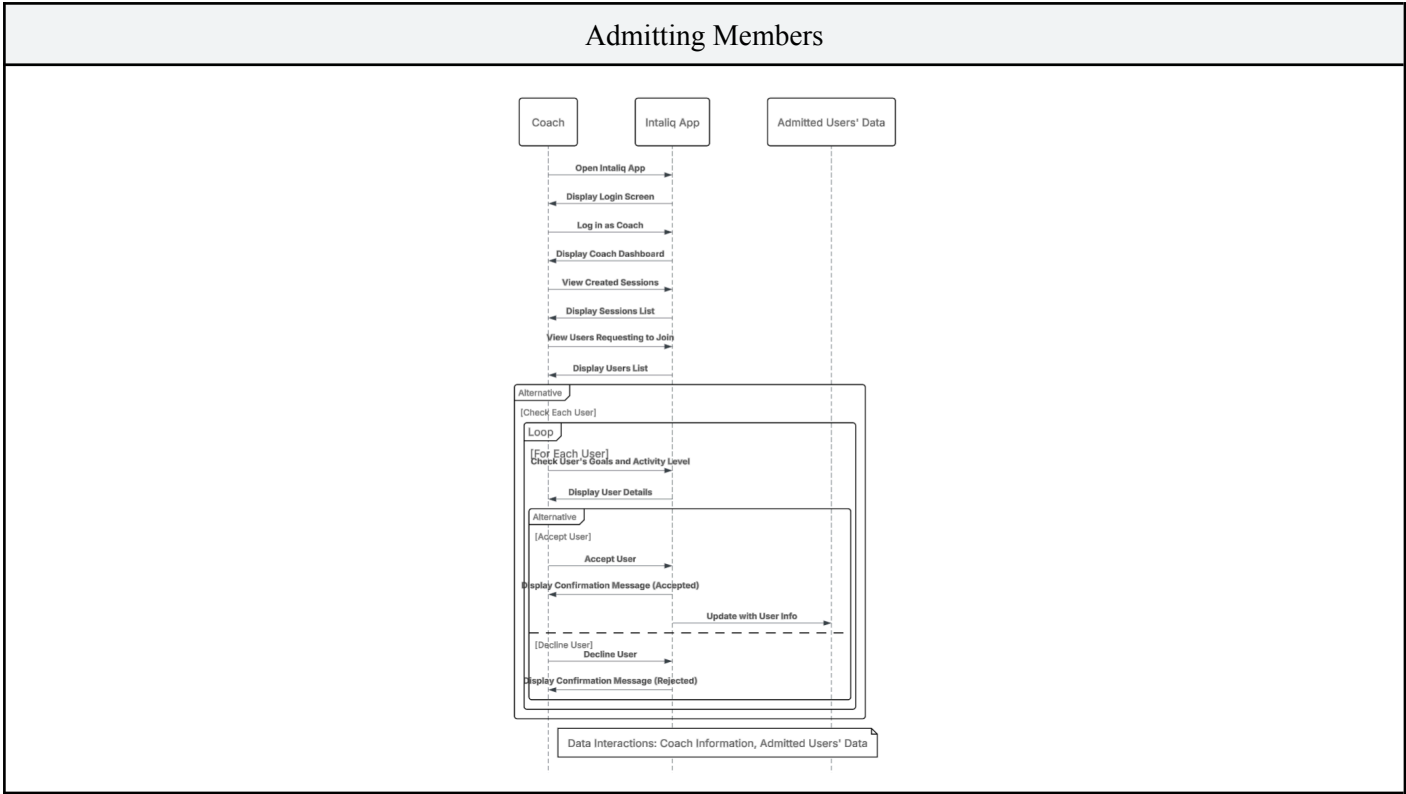


Booking a Group Session

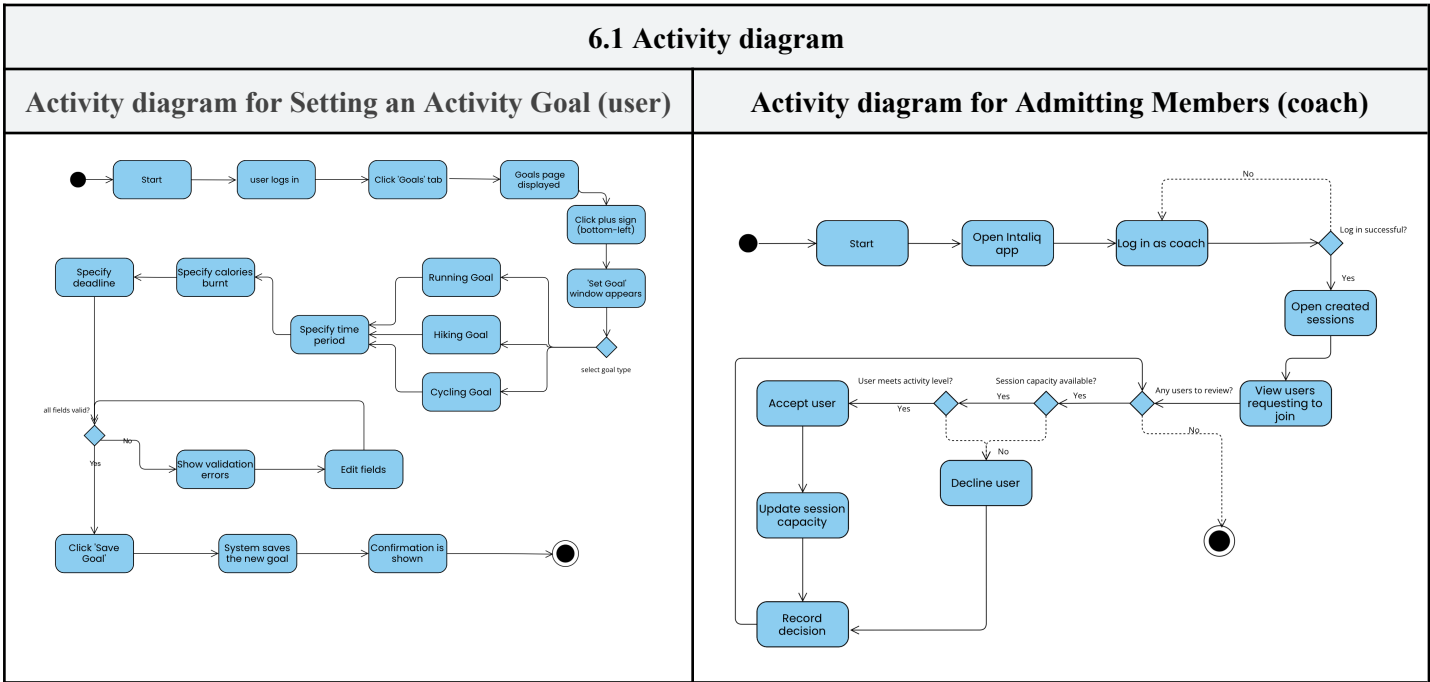


Creating a Group Session

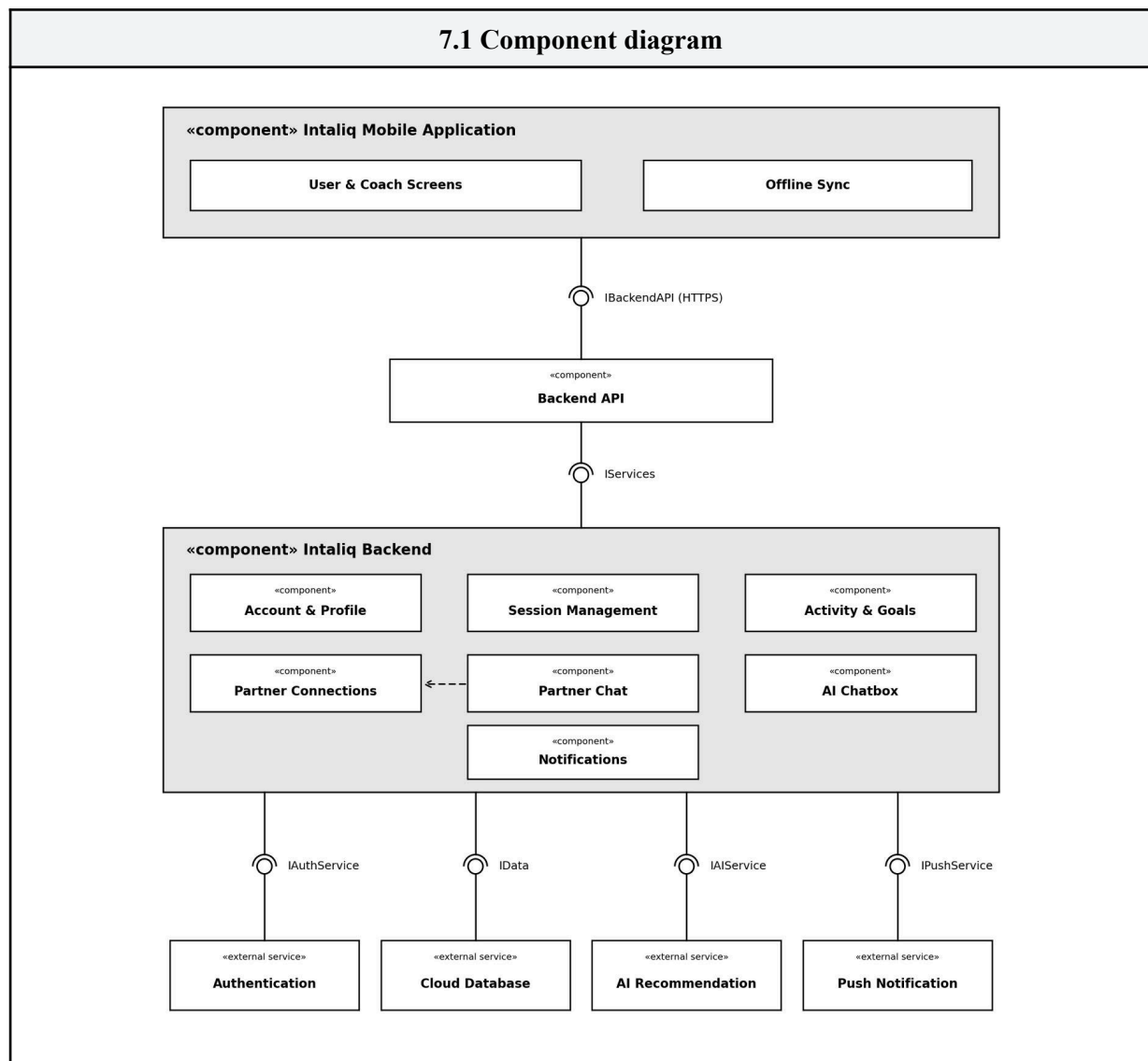




6. Process View



7. Development View



The component diagram displays the primary software components within Intaliq as well as their interfaces. Communication between the Mobile Application and the Backend API is via HTTPS. Within the Backend, there are multiple components that handle various tasks including Account & Profile Management, Sessions, Activities & Goals, Partner Connections, Partner Chat, AI Chat, and Notifications. The Backend interacts with external Authentication, Cloud Database, AI Recommendation, and Push Notification Services. Partner Chat is only accessible after a Partner Connection has been established, whereas Offline Sync stores all actions temporarily until connected back to the Internet.

8. Physical View

8.1 Deployment diagram

